NAME: ____________________ **PERIOD:** ________ **DATE:** ____________

Which Way Does the Condition Point?

AP Statistics · Unit 2 · Topic 2.6 · B: 2026-10-26 · E: 2026-11-13

[STAR] TODAY'S PROBLEM

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

Mina: "The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it."

Luis: "The base rate matters; the denominator should be everyone who tested positive."

Screening results (counts)

Status	Pos.	Neg.	Total
Condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

FIRST TAKE — before the video (5 min)

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

[BOOK] THE GIVEN EVENT SETS THE DENOMINATOR (keep on the page)

$P(A \mid B)$ means the probability of A **given** B. Restrict attention to the outcomes in B: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{\text{count in both } A \text{ and } B}{\text{count in } B}$. In general, $P(A \mid B) \neq P(B \mid A)$ because the denominators are different. **Sensitivity** is $P(\text{positive} \mid \text{condition})$. The **false-positive rate** is $P(\text{positive} \mid \text{no condition})$.

Finish after the video:

DOK 1 **(a)** For $P(\text{condition} \mid \text{positive})$, identify the numerator count and the denominator count from the table.

DOK 2 **(b)** Compute $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Show the count fraction for each and convert each to a percent.

DOK 3 **(c)** Decide whether Mina's claim is warranted. Cite the 45 positive results among students with the condition and the 95 positive results among students without it, explain how those counts determine the positive-test group, and diagnose the confusion between $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$.

Frame: The 90% describes $P($____________ $\mid$ ____________$)$, but the claim asks for $P($____________ $\mid$ ____________$)$.

Frame: Among the ________ positive results, ________ are true positives and ________ are false positives, so ____________________.

EXIT — turn this in

One thing the video changed about my first take: ______________________________